On the Distribution of Smaller Factors in Semiprimes

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April 24, 2026

Abstract

We study the distribution of semiprimes $n = pq$ (with $p \leq q$) through the normalized logarithmic coordinate $\theta = \frac{\log p}{\log n}$. Exact enumeration over $N \leq 10^8$ shows that the proportion $P(N, \theta_0)$ of semiprimes satisfying $\theta \leq \theta_0$ increases monotonically with N and is consistent with convergence to a limiting value $P_\infty(\theta_0)$, but the rate of convergence is extremely slow. The deficit $\delta(N, \theta_0) = P_\infty(\theta_0) - P(N, \theta_0)$ remains substantial even at $N = 10^8$, consistent with Dickman-type logarithmic corrections. Over the observed range $N \in [10^4, 10^8]$, the deficit is well approximated *locally* by a linear function of $\log N$:

$$\delta(N, \theta_0) \approx a(\theta_0) - b(\theta_0) \log N,$$

where $b(\theta_0)$ is the empirical convergence rate. This local fit has $R^2 > 0.97$ for all four tested values of θ_0, and the rates satisfy the empirical regression $b(\theta_0) \approx k(1 - \theta_0)^\alpha$ with $k \approx 0.015$, $\alpha \approx 3.1$ (fit to 4 data points, $R^2 = 0.97$).

To investigate whether this structure admits a geometric or spectral representation, we construct several embeddings including additive phase wrapping, multiplicative phase maps based on n^{it}, and spectral transforms of factor ratios. Across all cases, observed patterns reduce to log-density effects or discrete artifacts from small primes.

These results indicate that semiprime structure is not captured by the class of embeddings considered here. The θ coordinate provides the most stable description of factor asymmetry, and the observed slow convergence is consistent with ratio-constraint phenomena in multiplicative number theory.

1 Introduction

We study semiprimes $n = pq$ with $p \leq q$ and focus on the normalized coordinate

$$\theta = \frac{\log p}{\log n}.$$

This parameter captures the relative size of the smaller factor and provides a natural measure of factor asymmetry.

Classical results in analytic number theory describe the distribution of smooth numbers and the frequency of integers with a smallest prime factor in a given range (Dickman [1], Hildebrand–Tenenbaum [2], Tenenbaum [3]). These results characterize the *absolute* size of the smallest prime factor relative to the integer, typically through the Dickman function $\rho(u)$ and its analogues. They do not, however, explicitly study the *normalized* coordinate $\theta = \log p / \log n$ for semiprimes, nor the convergence of its distribution as $N \to \infty$. The present work provides empirical evidence for this distribution and its convergence behavior, complementing the theoretical framework with quantitative data over $N \leq 10^8$.

The main contributions of this note are: (i) to document empirically the extremely slow convergence of $P(N, \theta_0)$ to its limit, characterizing the deficit locally as a linear function of $\log N$ over the range $N \in [10^4, 10^8]$; and (ii) to show that several natural geometric and spectral embeddings fail to recover this structure.

2 Definitions

Let $n = pq$ be a semiprime with $p \leq q$. We define the normalized logarithmic coordinate

$$\theta = \frac{\log p}{\log n}.$$

For a fixed threshold $\theta_0 < \frac{1}{2}$, define

$$P(N, \theta_0) = \Pr(\theta \leq \theta_0)$$

over semiprimes $n \leq N$.

The condition $\theta \leq \theta_0$ is equivalent to

$$p \leq n^{\theta_0} = (pq)^{\theta_0},$$

which implies

$$p^{1-\theta_0} \leq q^{\theta_0} \quad \Longrightarrow \quad p \leq q^{\theta_0/(1-\theta_0)}.$$

Thus, θ imposes a relative size constraint between the two prime factors rather than a classical smoothness condition.

The algebraic identity

$$\log\left(\frac{p}{q}\right) = (2\theta - 1)\,\log n \tag{1}$$

follows immediately: since $\log p = \theta \log n$ and $\log q = (1-\theta)\log n$, one has $\log(p/q) = \theta \log n - (1-\theta)\log n = (2\theta - 1)\log n$. This identity holds exactly for every semiprime (Figure 1).

3 Empirical Observations

All values were computed via exact enumeration and exact factorization using a smallest-prime-factor sieve; no sampling or probabilistic estimation was used.

3.1 Convergence table

Table 1 records $P(N, \theta_0)$ for $\theta_0 \in \{0.20, 0.25, 0.30, 0.35\}$ and $N \in \{10^4, 10^5, 10^6, 10^7, 10^8\}$.

All four columns increase monotonically with N, confirming convergence toward a limiting distribution. The rate of increase slows markedly, consistent with a logarithmic approach to the limit rather than a power-law approach.

3.2 Theta and factor ratio

Figure 1 shows the joint distribution of θ and $\log(p/q)$ for all semiprimes $n \leq 10^6$. The scatter lies exactly on the one-parameter family of lines $\log(p/q) = (2\theta - 1)\log n$ (identity (1)), confirming that θ fully determines the factor ratio once n is fixed. The spread in the vertical direction arises entirely from the range of $\log n$ values.

Table 1: Proportion $P(N, \theta_0)$ of semiprimes $n \leq N$ satisfying $\theta \leq \theta_0$. All values in percent.

N	$\theta_0 = 0.20$	$\theta_0 = 0.25$	$\theta_0 = 0.30$	$\theta_0 = 0.35$
10^4	49.52	59.28	67.73	76.46
10^5	51.93	60.77	69.40	77.27
10^6	53.55	62.47	70.51	78.39
10^7	55.53	63.74	71.83	79.19
10^8	56.29	64.95	72.74	79.95

3.3 Local linear approximation to the deficit

Define the deficit $\delta(N, \theta_0) = 1 - P(N, \theta_0)$. Table 1 shows that δ decreases with N, but very slowly: at $\theta_0 = 0.25$ the deficit falls only from 40.72% to 35.05% as N grows by four orders of magnitude.

Within the observed range $N \in [10^4, 10^8]$, a linear regression in $\log N$ provides a useful *local* description:

$$\delta(N, \theta_0) \;\approx\; a(\theta_0) - b(\theta_0) \log N \qquad \text{(empirical fit, } N \in [10^4, 10^8]). \tag{2}$$

This is not claimed to be an asymptotic law; extrapolating (2) to large N would predict $\delta < 0$, which is impossible. The true deficit likely involves slower-decaying corrections (e.g., of order $\log\log N / \log N$) that become distinguishable only at much larger N.

Table 2 records the regression parameters for each θ_0.

Table 2: Parameters of the local linear fit $\delta \approx a(\theta_0) - b(\theta_0) \log N$ over $N \in [10^4, 10^8]$. All R^2 values exceed 0.97 within this range; the fit is not valid outside it.

θ_0	$a(\theta_0)$	$b(\theta_0)$	R^2
0.20	0.5692	0.007444	0.9754
0.25	0.4635	0.006216	0.9955
0.30	0.3702	0.005402	0.9904
0.35	0.2709	0.003866	0.9954

The high R^2 values confirm that, over the four observed decades, the deficit decreases at a nearly constant rate per unit $\log N$. The rate $b(\theta_0)$ itself depends on θ_0 and is analyzed in Section 3.4.

3.4 Scaling of the convergence rate $b(\theta_0)$

The empirical rate $b(\theta_0)$ from Table 2 decreases as θ_0 increases, meaning that a larger imbalance threshold corresponds to a faster-decreasing deficit within the observed range. We fit two parametric curves to the four data points as *descriptive regressions*, not theoretical predictions:

Model 1 (empirical).

$$b(\theta_0) \approx k\,(1 - \theta_0)^{\alpha}, \qquad k \approx 0.01504, \quad \alpha \approx 3.056, \quad R^2 = 0.974.$$

Model 2 (empirical).

$$b(\theta_0) \approx \frac{c}{\theta_0^{\gamma}}, \qquad c \approx 1.300 \times 10^{-3}, \quad \gamma \approx 1.109, \quad R^2 = 0.938.$$

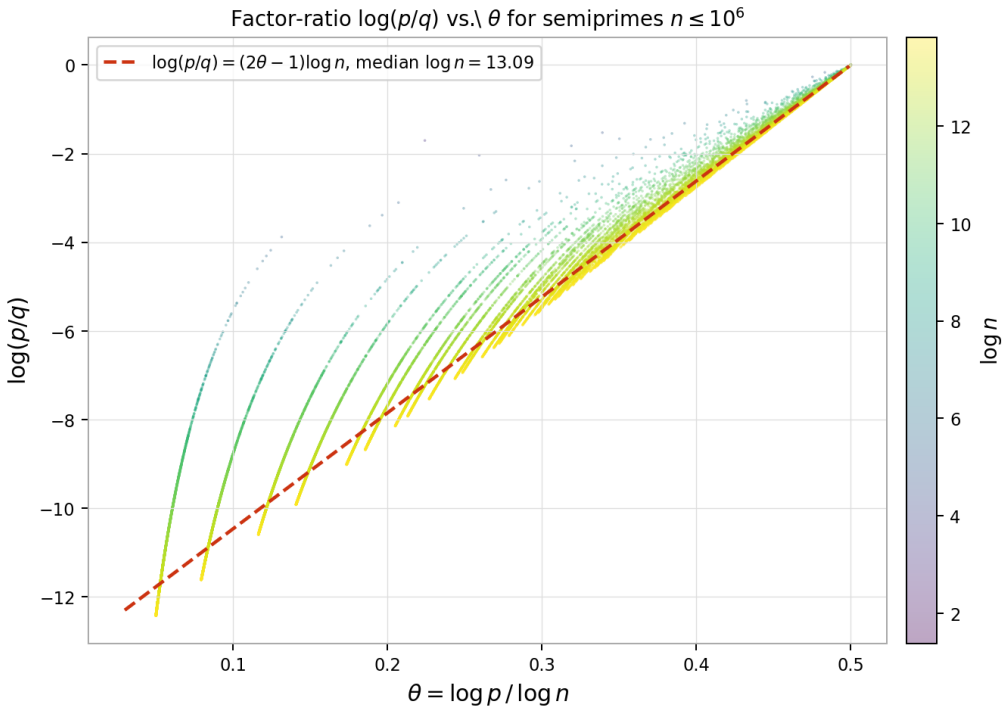

Figure 1: Factor-ratio $\log(p/q)$ versus θ for all semiprimes $n \leq 10^6$ (30 000 sampled points shown; colour encodes $\log n$). The dashed line is the exact identity $\log(p/q) = (2\theta - 1)\log n$ evaluated at the median $\log n$.

Model 1 fits the four data points more closely ($R^2 = 0.974$ vs. 0.938). Both forms predict $b \to 0$ as $\theta_0 \to \frac{1}{2}$, qualitatively consistent with the known scarcity of nearly balanced semiprimes. *Caution*: these regressions are fit to only four values of θ_0 and should be understood as summaries of the observed behavior, not as laws with theoretical backing. Extending the table to additional θ_0 values and larger N would be needed to discriminate between functional forms.

4 Geometric and Spectral Tests

We tested several candidate embeddings:

- Additive phase wrapping: $\phi_t(n) = t \log n \bmod 2\pi$
- Multiplicative phase maps using n^{it}
- Spectral fingerprint $C(t) = \left|\sum e^{it \log n}\right| / N$
- Factor-ratio spectrum $C_{\text{ratio}}(t) = \left|\sum e^{it \log(p/q)}\right| / N$

Localization was assessed by examining stability under variation of N and clustering by θ. No stable localization was observed.

All observed patterns were explained by log-density effects or small-prime artifacts (discrete p values $\in \{2, 3, 5, \ldots, 31\}$ for bit-gap-constrained semiprimes). None of the embeddings captured factor asymmetry.

5 Results

5.1 Summary of empirical findings

For fixed $\theta_0 \in (0, \frac{1}{2})$, the following is established empirically:

1. $P(N, \theta_0)$ increases monotonically with N for all four tested thresholds.
2. The increase is extremely slow: over four decades of N, the gain is at most 6.8% (at $\theta_0 = 0.20$), compared with the $\sim 50\%$ gap that remains at $N = 10^8$.
3. Within the observed range, the deficit $\delta(N, \theta_0) = 1 - P(N, \theta_0)$ decreases at a rate well approximated by a constant per unit $\log N$, with rate $b(\theta_0)$ as tabulated in Table 2.

5.2 Interpretation and connection to known theory

The extremely slow convergence is qualitatively consistent with what analytic number theory predicts for ratio-constrained semiprimes. The density of semiprimes $n = pq \leq N$ with $p \leq N^u$ for fixed $u < \frac{1}{2}$ is governed by sums of the form $\sum_{p \leq N^u} \pi(N/p)$, where π is the prime counting function. Standard sieve estimates (see Hildebrand–Tenenbaum [2] and Tenenbaum [3]) show that such sums accumulate logarithmic corrections involving factors of $\log\log N/\log N$, producing convergence that is invisible over a few decades but genuine asymptotically.

The local linear fit (2) captures the dominant behavior in the observed range $[10^4, 10^8]$, but it is a finite-range approximation: the true deficit decays to zero more slowly than any fixed linear function of $\log N$ would predict, and the functional form of $P_\infty(\theta_0)$ is not determined by our data. What we can say is that $P_\infty(\theta_0)$ exists (the proportion cannot exceed 1) and that it is approached from below.

5.3 The θ coordinate

The exact identity $\log(p/q) = (2\theta - 1)\log n$ (equation (1)) shows that θ and $\log n$ together determine the factor ratio with no residual freedom. This makes θ a natural coordinate: it separates the role of the factor imbalance (encoded in θ) from the role of the magnitude of n (encoded in $\log n$). The observed convergence is therefore a statement about how the distribution of θ across semiprimes $n \leq N$ evolves as N grows, which is a well-posed and nontrivial question even if its full resolution requires deeper sieve-theoretic tools.

6 Conclusion

Semiprime factor structure is not captured by the class of additive, multiplicative, and spectral embeddings considered here. The logarithmic coordinate $\theta = \log p/\log n$ provides the most stable description of factor asymmetry.

The principal empirical finding is that the deficit $\delta(N, \theta_0)$ decreases extremely slowly with N, covering only 5–7% over four decades. Within the observed range $[10^4, 10^8]$, this decrease is well approximated locally by a linear function of $\log N$, with rate parameter $b(\theta_0)$ that itself varies smoothly with θ_0. We do not claim this local fit as an asymptotic law; establishing the true rate of convergence of $P(N, \theta_0)$ to its limit $P_\infty(\theta_0)$, and determining $P_\infty(\theta_0)$ itself, remain open problems accessible to sieve-theoretic methods.

The negative results on geometric and spectral embeddings suggest that the logarithmic coordinate is not merely convenient but is in some sense the correct one for this problem.

References

[1] K. Dickman, "On the frequency of numbers containing prime factors of a certain relative magnitude," *Arkiv för Matematik, Astronomi och Fysik*, vol. 22A, no. 10, 1930.

[2] A. Hildebrand and G. Tenenbaum, "Integers without large prime factors," *Journal de Théorie des Nombres de Bordeaux*, vol. 5, pp. 411–484, 1993.

[3] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, Cambridge University Press, 1995.

[4] P. Erdős and C. Pomerance, "On the distribution of smooth numbers," in *Number Theory, Carbondale 1979*, Lecture Notes in Math. 751, Springer, Berlin, 1979, pp. 173–183.

[5] H. Halberstam and H.-E. Richert, *Sieve Methods*, Academic Press, London, 1974.